

SIH SOCIAL MEDIA ANALYTICS & THREAT PRIORITIZATION

Consolidated Engineering & Research Meeting Proceedings (Meetings 1 to 4)

Project Scope: AI-Powered Instagram & Twitter Intelligence, Sentiment Pipeline & Heuristic Risk Scoring

Date Range: 27/08/2026 – 30/08/2026

Architecture: Python / Pandas, TF-IDF + Logistic Regression, VADER NLP, Human-in-the-Loop (HITL)

Status: Prototype Framework Complete

MEETING 1

SIH Problem Statement & Initial Strategy

Date: 27/08/2026

1. Problem Understanding & Core Vision

The core objective is to architect an AI-based social media analytics engine converting unstructured social data into actionable intelligence. The system focuses on tracking topic shifts, measuring audience sentiment (Positive / Negative / Neutral), mapping demographic engagement patterns, and leveraging temporal analysis (timestamps) to monitor evolving narrative dynamics.

2. Internal Prototype Strategy

Rather than building a full-scale multi-platform deployment immediately, the team agreed on an *in-depth lean prototype* approach prioritizing data integrity, robust cleaning pipelines, and verifiable statistical insights over broad superficial coverage.

3. Datasets & Baseline Pipeline

- Twitter Corpus:** Avengers: Endgame dataset for temporal activity modeling, amplification tracking, and user interaction graphs.
- Instagram Corpus:** Multi-variable content performance dataset (29,999 records) assessing media formats, follower growth, reach, and interaction intensity.
- Tech Stack Scope:** CSV Ingestion → Pandas Pipeline & Structuring → Distinct Platform Feature Engineering → Metric Dashboards. Complex cloud streaming & black-box models are deferred to subsequent phases.

MEETING 2

Platform Deep-Dive: Engagement & Conversation Dynamics

Date: 28/08/2026

1. Instagram Engagement & Audience Response Framework

Evaluated 29,999 records across 23 distinct metrics (account, content, timing, reach, impressions, follower delta). Interactions were segmented into distinct intent signals: *Likes (Attention)*, *Comments (Conversation)*, *Shares (Propagation)*, and *Saves (Perceived Long-Term Value)*.

Standardized Engagement Formula:

$$Engagement\ Rate = [(Likes + Comments + Shares + Saves) / Followers] \times 100$$

2. Twitter Conversation Flow & Baseline Sentiment (VADER)

- Retweet Dominance:** The conversation exhibited a massive retweet ratio of ~0.90, demonstrating that viral momentum is driven primarily by amplification rather than original content generation.
- Total Tweet Sentiment Distribution:** Positive: 52.94% | Neutral: 38.42% | Negative: 8.64%.
- Original Tweets Breakdown:** Neutral: 46.56% | Positive: 38.96% | Negative: 14.48%.

Rationale

1. Instagram Feature Timing & Category Insights

- **Post Hour Priority:** Posting time dictates immediate algorithmic feed distribution. Peak performing hours: **03:00, 08:00, 02:00, 17:00, 20:00**. Highest weekdays: **Tuesday (0.04265) & Sunday (0.04248)**.
- **Top Performing Niches:** Music (0.0428), Fitness (0.0427), Fashion (0.0426).
- **Propagation Metric:** *Shares* selected over likes/comments as the direct measure of spread velocity for threat modeling.

2. Twitter Supervised NLP Pipeline & Data Leakage Prevention

Because raw tweets lacked ground truth sentiment labels, 2 external labeled corpora (~74k tweets) were merged and cleaned to 57,178 training records. A severe **52% train-test data leakage** was identified and rectified, isolating a clean, unseen 398-tweet benchmark test set.

92.96%

INITIAL TF-IDF MODEL

98.74%

TUNED LOGISTIC REG.

5 / 398

CONTEXTUAL EDGE CASES

Human-in-the-Loop (HITL) 100% Accuracy Roadmap: The remaining 5 misclassifications stemmed from deep context, sarcasm, and ambiguous news reporting. The system integrates an automated confidence threshold: predictions with low confidence scores are routed to human analysts, ensuring zero false positives in operational deployments.

Roadmap

1. VADER Sentiment Corpus (29,999 Records)

13,804 (46.0%)

POSITIVE POSTS

7,648 (25.5%)

NEUTRAL POSTS

8,547 (28.5%)

NEGATIVE POSTS

2. Why Negative Sentiment Alone is Insufficient

A crucial empirical finding showed that average engagement and reach across positive, neutral, and negative posts are virtually identical. Hostile posts without distribution pose low systemic danger, while rapidly amplifying neutral or ambiguous content may trigger unrest. Sentiment must therefore be mathematically coupled with propagation metrics.

3. Trend Anomaly & High-Share Negative Filtering (90th Percentile)

- **Volume Normalization:** Aggregated weekly via `groupby(['date', 'sentiment']).size().unstack()` . Identified a significant surge in negative sentiment during **Week 4 (~35%)** over baseline (~28%).
- **Top 10% Risk Quarantine:** Negative posts filtered at `quantile(0.90)` on shares. Benchmark post: 516 shares, 46,738 reach, 0.2398 engagement, sentiment score -0.8655.

4. Prototype Heuristic Risk Scoring Formula

$$Risk\ Score = [(0.30 \times Neg_Score) + (0.25 \times Share_Score) + (0.25 \times Reach_Score) + (0.20 \times Eng_Score)] \times 100$$

Feature Component	Weight	Operational Representation	Sample Weighted Output
Negative Sentiment	30%	Normalized polarity strength [0, 1]	$0.80 \times 0.30 = 0.240$

Feature Component	Weight	Operational Representation	Sample Weighted Output
Share Magnitude	25%	Relative viral spread velocity	$0.70 \times 0.25 = 0.175$
Reach Magnitude	25%	Audience exposure breadth	$0.90 \times 0.25 = 0.225$
Engagement Rate	20%	Interaction density (likes + comments)	$0.60 \times 0.20 = 0.120$
Total Combined Composite Score (Raw: 0.760×100)			76 / 100 (HIGH RISK)

Triage Range	Classification	Operational Action
0 – 39	Low Risk	Standard baseline metrics; passive logging.
40 – 69	Medium Risk	Elevated virality / negative tone; periodic queue check.
70 – 100	High Risk	Multi-signal convergence; immediate analyst escalation.

5. Future Supervised Machine Learning Roadmap

- Ground Truth Definition:** Establish precise threat taxonomies verified by domain subject-matter experts.
- Deep Feature Engineering:** Combine BERT/RobERTa contextual embeddings with graph network centrality and propagation decay curves.
- Model Ensembling:** Train Supervised Classifiers (Logistic Regression, Random Forest, XGBoost) calibrated against multi-signal labels.
- Integrated Triage Dashboard:** Real-time visualization interface combining automated scoring with human reviewer verification.